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Fecha: Marzo 5, 2025

INSTRUCCIONES: Del siguiente banco de ejercicios, escoger tres y resolverlos.

Problemas

Problem 1:

Suppose we take 2 coins at random from a box that contains

- 4 coins of \$2
- 5 coins of \$1
- 8 coins of ¢50
- 3 coins of ¢20

Let $X = k_i$ be the winning gain after choosing the coins.

Revision for Solution

Notice the following, we are dealing with a sampling without replacement from a finite population (i.e population $N = 4 + 5 + 8 + 3 = 20$ coins), but the number of categories in our sampling space is 4 (i.e $c = 4$), and we have only two observations.

Lets consider the following Mass Probability Function (MPF): Multivariate Hypergeometric Distribution (Wikipedia, 2025b)^a:

$$P(X_1 = k_1, \dots, X_4 = k_4) = \begin{cases} \frac{\prod_{i=1}^4 \binom{K_i}{k_i}}{\binom{N}{n}} & k \in (Z_{0+})^4 : \forall i k_i \leq K_i, \sum_{i=1}^c k_i = n \\ 0 & \text{otherwise} \end{cases}$$

So defining our categories as the problem suggested:

- c_1 : 4 coins, $K_1 = \$2$
- c_2 : 5 coins, $K_2 = \$1$
- c_3 : 8 coins, $K_3 = ¢50$
- c_4 : 3 coins, $K_4 = ¢20$

As suggested by Wikipedia (2025b) there is no closed form for the CDF in the multivariate Hypergeometric Distribution, so to solve Subsection 1.1, we gonna have to directly find the CDF via point wise computation. However, we can find the Expectation for the

Multivariate Hypergeometric Distribution.

Before jumping into solving the expectation, let's discuss what we need to prove, if we decide to apply the definition of expectation directly to the PMF, we would encounter something like this:

$$\begin{aligned} E[X_i] &= E\left[\frac{\prod_{i=1}^4 \binom{K_i}{x_i}}{\binom{N}{n}}\right] \\ &= \sum_x x_i \left(\frac{\prod_{i=1}^4 \binom{K_i}{x_i}}{\binom{N}{n}}\right) \\ &= ?? \end{aligned}$$

Evidently, we encounter an issue, with finding the summation of the product of multiple binomials. But luckily there is a workaround for this issue, as suggested by Adhikari (2025), if we want to find the expectation for expected counts, then one can use an Indicator function I , defining each count with a respective indicator: $N = I_1 + I_2 + \dots + I_n$, since expectations are a linear operator then $E[I_A] = P(A)$. This saves us the trouble of having to actually compute the summation.

Proof.

Lets define the following Indicator function :

$$I_{ij} = \begin{cases} 1, & \text{if the } j\text{-th draw is of type } i \\ 0, & \text{otherwise} \end{cases}$$

Then the total number of type i coin drawn results in:

$$X_i = \sum_{j=1}^n I_{ij}$$

Consider the probability of the Indicator, given that the possible results for the indicators are mutually exclusive, then

$$P(I_{ij}) = P(I_{ij} = 1) + P(I_{ij} = 0)$$

Since the drawing is random without replacement from a total of N coins, with K_i different types i of coins, so the probability that any given draw is of type i results:

$$P(I_{ij} = 1) = \frac{K_i}{N}$$

By the definition of Expectation:

$$E[I_{ij}] = (1)(P(I_{ij})) + (0)(P(I_{ij})) = \frac{K_i}{N}$$

Then find the expectation of the total number of type i coin, which the equality holds by linearity of the expectation:

$$E[X_i] = E\left[\sum_{j=1}^n I_{ij}\right] = \sum_{j=1}^n E[I_{ij}] = n \frac{K_i}{N}$$

$$\therefore E[X_i] = n \frac{K_i}{N}$$

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^aAlthough Wikipedia is generally not considered an appropriate place for references; I would recommend that Wikipedia is a concise source to find general information about probability distributions.

Subsection 1.1

Determine $F(X)$ be the probability distribution function (CDF) of X

Revision for Solution

As suggested by Wikipedia (2025b), there is no closed form CDF for the multivariate Hypergeometric Distribution. Then we can determine the PMF for the exact probabilities, using the Multivariate Hyper Geometric distribution, and then by definition of CDF, find the CDF from the exact probabilities.

Recall that we have the following conditions:

Total number of coins $N = 4 + 5 + 8 + 3 = 20$

The number of observations $n = n_A + n_B + n_C + n_D = 2$

(For simplification purposes, we define A as K_1 , B as K_2 , C as K_3 , D as K_4 .)

Hence the total winning:

$$P(n_A, n_B, n_C, n_D) = \frac{\binom{4}{n_A} \binom{5}{n_B} \binom{8}{n_C} \binom{3}{n_D}}{\binom{20}{2}}$$

Then if X is the winning gain for each category, we then define:

$$X = 2n_A + 1n_B + 0.50n_C + 4n_D$$

The problem now is that we have to define the probabilities for every combination of two coins that the box has and its corresponding gain.

Consider the case where is the same type of coin:

$$\begin{aligned}P(A, A) &= \frac{\binom{4}{2}}{\binom{20}{2}} = \frac{6}{190} & X &= 2 + 2 = 4.00 \\P(B, B) &= \frac{\binom{5}{2}}{\binom{20}{2}} = \frac{10}{190} & X &= 1 + 1 = 2 \\P(C, C) &= \frac{28}{190} & X &= 0.50 + 0.50 = 1 \\P(D, D) &= \frac{3}{190} & X &= 0.20 + 0.20 = 0.40\end{aligned}$$

Well know we just have to do it for the combination where the coin is not of the same type.

$$\begin{aligned}P(A, B) &= \frac{\binom{4}{1}\binom{5}{1}}{\binom{20}{2}} = \frac{20}{190} & X &= 2 + 1 = 3 \\P(A, C) &= \frac{32}{190} & X &= 2 + 0.50 = 2.50 \\P(A, D) &= \frac{12}{190} & X &= 2 + 0.20 = 2.20 \\P(B, C) &= \frac{32}{190} & X &= 1 + 0.50 = 1.50 \\P(B, D) &= \frac{15}{190} & X &= 1 + 0.20 = 1.20 \\P(D, C) &= \frac{24}{190} & X &= 0.50 + 0.20 = 0.70\end{aligned}$$

Now organize from the smallest x to the largest x, and with that we obtain our PMF:

$$f(x) = \begin{cases} \frac{3}{190}, & x = 0.40 \\ \frac{24}{190}, & x = 0.70 \\ \frac{28}{190}, & x = 1.00 \\ \frac{15}{190}, & x = 1.20 \\ \frac{40}{190}, & x = 1.50 \\ \frac{10}{190}, & x = 2.00 \\ \frac{12}{190}, & x = 2.20 \\ \frac{32}{190}, & x = 2.50 \\ \frac{20}{190}, & x = 3.00 \\ \frac{6}{190}, & x = 4.00 \\ 0, & \text{otherwise} \end{cases}$$

Solution:

So from the PMF from above, we can construct our PMF simply as:

$$F(x) = \begin{cases} 0, & x < 0.40 \\ \frac{3}{190}, & 0.40 \leq x < 0.70 \\ \frac{27}{190}, & 0.70 \leq x < 1.00 \\ \frac{55}{190}, & 1.00 \leq x < 1.20 \\ \frac{70}{190}, & 1.20 \leq x < 1.50 \\ \frac{110}{190}, & 1.50 \leq x < 2.00 \\ \frac{120}{190}, & 2.00 \leq x < 2.20 \\ \frac{132}{190}, & 2.20 \leq x < 2.50 \\ \frac{164}{190}, & 2.50 \leq x < 3.00 \\ \frac{184}{190}, & 3.00 \leq x < 4.00 \\ 1, & 4.00 \leq x \end{cases}$$

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Subsection 1.2

Calculate $E[x_i]$

Solution:

As shown before, we have that $X \sim \text{MultivariateHypergeometric}(c, K_4, N)$, and we have the observations $n = 2$, where

$$\begin{aligned} c &= 4 \text{ types of coins} \\ K_i &= \sum_{i=1}^4 k_i = \$17.60 \\ N &= 20 \text{ coins} \end{aligned}$$

As proven in the revision for solution, we have that $E[X] = n \frac{K_i}{N}$, hence:

$$E[x_i] = 2 \frac{17.60}{20} = \$1.76$$

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Problem 2:

Suppose that the probability distribution function $F(x)$ has a random variable X is given by:

$$F(x) = \begin{cases} 0, & x < 0 \\ \frac{x}{2}, & 0 \leq x < 1 \\ \frac{x+1}{4}, & 1 \leq x < 3 \\ 1, & x \geq 3 \end{cases}$$

Subsection 2.1

$$P(\{X < 1\})$$

Solution:

By definition for the CDF of X, then we have that:

$$F(\{X < 1\}) = F(1) = \frac{1}{2}$$

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Subsection 2.2

$$P(\{X > 1\})$$

Solution:

By definition of the CDF, notice that we inequality is defined as $P(\{X < x\})$, so by A2, we have that

$$P(\{1 < X\}) = 1 - P(\{X < 1\}) = 1 - F(1) = \frac{1}{2}$$

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Subsection 2.3

$$P(\{\frac{1}{3} \leq X < \frac{5}{3}\})$$

Solution:

With this problem we can decompose the problem into two probabilities:

$$P\left(\frac{1}{3} \leq X\right) \text{ \& } P\left(X < \frac{5}{3}\right)$$

Do notice that $\frac{1}{3} < 1$ and $\frac{5}{3} < 3$, similarly to subsection 2.2, we can rewrite $P(\frac{1}{3} < X) = 1 - P(X < \frac{1}{3})$. Clearly this result in that

$$\begin{aligned} P\left(\left\{\frac{1}{3} \leq X < \frac{5}{3}\right\}\right) &= \left(1 - P(X < \frac{1}{3})\right) - \left(1 - P(X \leq \frac{5}{3})\right) = (P(X < \frac{5}{3})) - (P(X \leq \frac{1}{3})) \\ &= F\left(\frac{5}{3}\right) - F\left(\frac{1}{3}\right) = \frac{\frac{5}{3} + 1}{4} - \frac{\frac{1}{3}}{2} = \frac{2}{3} - \frac{1}{6} = \frac{1}{2} \end{aligned}$$

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Problem 3:

In a tournament, one team plays against another three distinct teams. The probabilities of winning against each opponent:

- First team : $\frac{4}{5}$
- Second team : $\frac{13}{20}$
- Third team : $\frac{3}{10}$

Suppose that each winning probability is independent. So let X be the number of obtained victories.

Revision for Solution

One way to look at this problem is as if these matches are defined as Bernoulli trials, however, we are dealing with more than 2 trials and the probability of winning on each match differs, so the standard Bernoulli Distribution is not a proper fit. Nevertheless there exists the Poisson Binomial distribution, which is sort of like a generalized Bernoulli distribution, which takes the sum of the independent Bernoulli trials relaxing the condition that each trial is identically distributed.

Subsection 3.1

Determine the probability mass function of X

Solution:

The defined PMF for the Poisson Binomial Distribution (Wikipedia, 2025c):

$$P(X = x) = \begin{cases} \sum_{A \in F_k} \prod_{i \in A} p_i \prod_{j \in A^c} (1 - p_j), & x \in \{0, \dots, n\} \\ 0 & \text{otherwise} \end{cases}$$

Here X can be any of the following observations:

- $x_0 = 0$ Victories, with prob p_0 : $\frac{49}{1000}$
- $x_1 = 1$ Victories, with prob p_1 : $\frac{4}{5}$
- $x_2 = 2$ Victories, with prob p_2 : $\frac{13}{20}$
- $x_3 = 3$ Victories, with prob p_3 : $\frac{3}{10}$

To obtain p_0 , one approach is to $P(X = 0)$ by directly computing using the defined PMF above then:

- $q_1 = 1 - p_1 = \frac{1}{5}$
- $q_2 = 1 - p_2 = \frac{7}{20}$
- $q_3 = 1 - p_3 = \frac{7}{10}$
- $P(X = 0) = (q_1)(q_2)(q_3) = (1 - p_1)(1 - p_2)(1 - p_3) = \frac{1}{5} \times \frac{7}{20} \times \frac{7}{10} = \frac{49}{1000}$

Although notice there is an error with the defined probabilities, and this can be verified because of Axiom 2. Notice the following:

$$\frac{49}{1000} + \frac{4}{5} + \frac{13}{20} + \frac{3}{10} = \frac{1799}{1000} \neq 1$$

Evidently, to obtain the required PMF for this distribution, it is required that the product of the sample space for the given Bernoulli trials results in 1. Which means that to obtain the probabilities for x_1, x_2 , & x_3 we need to apply the calculations given by the Poisson Binomial Distribution, which is first to find the product for each Bernoulli trial, and then sum these trials. Hence: $P(X = 1)$ (they win exactly only one victory):

$$\begin{aligned} \text{Win against only Team 1: } (p_1)(q_2)(q_3) &= \frac{4}{5} \times \frac{7}{20} \times \frac{7}{10} = \frac{196}{1000} \\ \text{Win against only Team 2: } (q_1)(p_2)(q_3) &= \frac{1}{5} \times \frac{13}{20} \times \frac{7}{10} = \frac{91}{1000} \\ \text{Win against only Team 3: } (q_1)(q_2)(p_3) &= \frac{1}{5} \times \frac{7}{20} \times \frac{3}{10} = \frac{21}{1000} \end{aligned}$$

Now we add up all these probabilities, just as the PMF for the Poisson Binomial suggests we do.

$$P(X = 1) = \frac{196}{1000} + \frac{91}{1000} + \frac{21}{1000} = \frac{308}{1000} = \frac{77}{250}$$

Now we have to do the same process for $P(X=2)$ and $P(X=3)$, but could there be a faster approach to solve the problem above, so that we don't need to keep calculating the Probabilities one by one?

Notice the kind of pattern produced by defining the product of the probabilities p_i and their corresponding complementary probabilities q_i , then lets define the following:

$$\begin{aligned} M(t_1) &= q_1 + p_1 t = \frac{1}{5} + \frac{4}{5} t \\ M(t_2) &= q_2 + p_2 t = \frac{7}{20} + \frac{13}{20} t \\ M(t_3) &= q_3 + p_3 t = \frac{7}{10} + \frac{3}{10} t \end{aligned}$$

Now because we know that each Bernoulli trial is independent, we can multiply these $M(t_i)$:

$$M(t_1)M(t_2)M(t_3) = \left(\frac{1}{5} + \frac{4}{5}t\right)\left(\frac{7}{20} + \frac{13}{20}t\right)\left(\frac{7}{10} + \frac{3}{10}t\right)$$

$$\Rightarrow (\frac{7}{100} + \frac{41}{100}t + \frac{13}{25}t^2)(\frac{7}{10} + \frac{3}{10}t)$$

Lets continue finishing the product

$$\Rightarrow \frac{49}{1000} + \frac{21}{1000}t + \frac{287}{1000}t + \frac{123}{1000}t^2 + \frac{91}{250}t^2 + \frac{39}{250}t^3$$

Lets add all the probabilities with the same polynomial.

$$\Rightarrow \frac{49}{1000} + \frac{77}{250}t + \frac{487}{1000}t^2 + \frac{39}{250}t^3$$

To the person reading this, hello Ariela, if you reading this and asking whats going on, perhaps you should check your notes on Mathematical Statistics. Because what just happened is I applied the Moment Generating Function for the Poisson Binomial distribution (again check Wikipedia, 2025c). Which resulted in a polynomial with degree 3, which makes sense given that we only have three Bernoulli trials. And surprise, each coefficient is the $P(X=x)$. Hence our PMF for the Bernoulli trials are:

$$f(x) = \begin{cases} \frac{49}{1000}, & x = 0 \\ \frac{77}{250}, & x = 1 \\ \frac{487}{1000}, & x = 2 \\ \frac{39}{250}, & x = 3 \\ 0, & \text{otherwise} \end{cases}$$

And do notice that

$$\frac{49}{1000} + \frac{77}{250} + \frac{487}{1000} + \frac{39}{250} = 1$$

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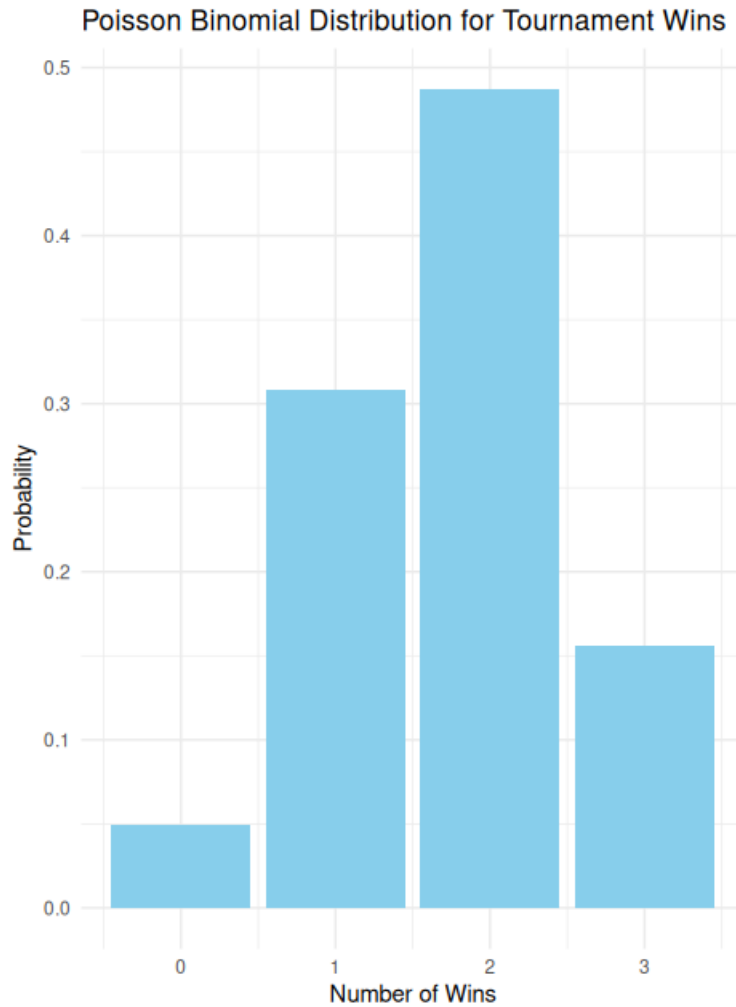


Figure 1: R ver. 4.3.3

Subsection 3.2

Determine the probability distribution function of X

Revision for Solution

As suggested by Wikipedia (2025c) the PMF has a closed-form CDF. So to determine the CDF for a PMF, there are two approaches as suggested by Tang and Tang (2023) either through recursive formulas or discrete Fourier Analysis. Although note that both approaches require some degree of computational algorithm, making it unfeasible to do it without the aid of a computer, which unfortunately isn't the aim of this assignment. Hence we gonna have to directly compute the CDF for the problem without using the Poisson binomial distribution. But for the curios mind the CDF for the Poisson Binomial distribution:

$$P(K \leq k) = \sum_{l=0}^k \sum_{A \in F_l} \Pi_{i \in A}$$

Solution:

Consider the PMF found before, then recall that $F(x) = P(X \leq x)$, then:

$$F(X) = \begin{cases} 0, & \text{if } x < 0 \\ \frac{49}{1000} & \text{if } 0 \leq x < 1 \\ \frac{357}{1000} & \text{if } 1 \leq x < 2 \\ \frac{211}{250} & \text{if } 2 \leq x < 3 \\ 1 & \text{if } x \geq 3 \end{cases}$$

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Subsection 3.3

Determine the expected value of X

Revision for Solution

As usual, one can directly compute the expectation of the PMF, but as a general advice, that is usually not recommended. As you already notice the PMF for the Poisson-Binomial is rather eccentric, given that there is two summation and a product on top of the summations. So just like any other discipline in mathematics, one must determine the result without directly using its definition. Notice that the distribution is simply the generalized version of the Bernoulli, which the shape of the distribution is an indicator to suggest whether they belong in the same family of distributions. Therefore the expectation for the Poisson Binomial is simply the $\sum_{i=1}^n E[Y]$ where $Y \sim \text{Bernoulli}(p)$, hence:

Proof. Let $Y \sim \text{Bernoulli}(p)$ then,

$$E[Y] = E[p^y(1-p)^{1-y}] = E\left[\sum_y xP(X=y)\right]$$

Since Y only has support in $\{0,1\}$

$$= P(Y=1)(1) + P(Y=0)0$$

Since each event is independent, then $P(Y=1) = p$ and $P(Y=0) = (1-p)$

$$= p(1) + (1-p)(0) = p$$

■

$$\text{Therefore } E[X] = \sum_{i=1}^n E[Y] = \sum_{i=1}^n p_i$$

Solution:

As proven before, $E[X] = \sum_{i=1}^n E[Y] = \sum_{i=1}^n p_i$, if you are in doubt that similar shape results in identical properties (this is usually called Location-Scale Family), you can familiarize yourself with the concept Location-Scale Family in Siegrist (2025).

Hence the result for the expectation $E[X] = \sum_{i=1}^3 p_i = \frac{4}{5} + \frac{13}{20} + \frac{3}{10} = 1.75$ victories ■

Problem 4:

Five unique numbers are distributed between five players from 1 to 5. Every time two players compare their numbers, whoever has the largest number is declared the winner. Initially player 1 and player 2 compare their numbers, then the winner compares their number with player 3, and we proceed consequentially.

A key observation is that for each round the victorious player is whichever the observed number is the highest. In particular player 1 will win the first round, if and only if they have a higher number than player 2, then player 1 will be victorious if and only if its number is higher than player 3 and player 2.

Let X be the number of times the player 1 resulted victorious.

Revision for Solution

Notice that for each round, a new player enters the encounter, resulting in that we have no replacement. Because there is no replacement, we know that the end stops at the 5th round. Also notice that each round would correspond to a Bernoulli trial (i.e. p = winner, q = not winner = $1-p$). Finally, each round would be independent of other rounds, since the value of the number is not modified for each round that is proceeded. We indeed have another Poisson Binomial Distribution.

Well the reasoning above is technically wrong as suggested by ChatGPT (2025). Although the AI agreed with my idea that we dealing with no replacement and that the game stops at the 5th round; nevertheless, the part of the Bernoulli Trails and the independence of them had a disagreement with me. This is because the probability of player 1 winning on each round is not constant, as it depends on how many players they have played with. This should have been noticed due to the nature of the game being sequential. Evidently, each round is not independent of each other if they depend on the sequence of previous rounds. So using the Poisson Binomial Distribution is not a proper fit for this problem.

So considering that the problem has a nature of playing each round sequentially, then the obvious answer would be to look for a distribution who's probability mass function handles sequential events and has no replacement. Nevertheless, I couldn't find a distribution on the internet that would work in the established problem. So I asked again ChatGPT (2025), and it suggested based on Arnold et al. (1998) that the theory of records is the proper framework to model sequential events. It suggested to use the "record process" to model sequential events that have no replacement. Specifically that

we can model these events with probability:

$$\frac{1}{k+1}, \text{ after } k+1 \text{ observations}$$

Subsection 4.1

Determine the probability mass function for X.

Solution:

So from the above, we can model our PMF using sequential rounds. As suggested before each sequential round can be express using $\frac{1}{x+1}$, after $x+1$ observations. The reason as to why we start observing after $x+1$, is because we can guaranteed that player 1 won its first game after the first round passed. Nevertheless, because player one can lose on the first round, then our X must support the case where player 1 wins 0 games. Again, since the game only lasts until five rounds, then we can define $X\{0, 1, 2, 3, 4\}$, then if we start thinking on the probabilities we would have something like:

$P(X \geq 1)$	the prob. that p 1 has the largest number between p 1 and p 2	$\frac{1}{2}$
$P(X \geq 2)$	the ... among all players in round 3	$\frac{1}{3}$
$P(X \geq 3)$	the ... among all players in round 4	$\frac{1}{4}$
$P(X \geq 4)$	the ... among all players in round 4	$\frac{1}{5}$

Again, because the nature of the game is that each round it depends sequentially from the previous, it must result in that the exact probability of winning round j is the difference between the previous round from round j, and the upcoming round of j. I.e.

$$P(X = j) = P(X \geq j) - P(X \geq j + 1)$$

With the above we can construct the probabilities for the PMF:

$$P(X = 0) = P(X \geq 0) - P(X \geq 1)$$

Notice that difference is just as shown in subsection 2.3. Also $P(X \geq 0)$ is equivalent to all the rounds in the game, which must be equal to 1 (by A2); meanwhile, $P(X \geq 1)$ is the defined probability from above.

$$P(X = 0) = 1 - \frac{1}{2} = \frac{1}{2}$$

Meanwhile for the other $j = 1, 2, 3, 4$:

$$\begin{aligned} P(X = 1) &= P(X \geq 1) - P(X \geq 2) = \frac{1}{2} - \frac{1}{3} = \frac{1}{6} \\ P(X = 2) &= P(X \geq 2) - P(X \geq 3) = \frac{1}{3} - \frac{1}{4} = \frac{1}{12} \\ P(X = 3) &= P(X \geq 3) - P(X \geq 4) = \frac{1}{4} - \frac{1}{5} = \frac{1}{20} \\ P(X = 4) &= P(X \geq 4) - P(X \geq 5) = \frac{1}{5} - 0 = \frac{1}{5} \end{aligned}$$

Hence our PMF:

$$f(x) = \begin{cases} \frac{1}{2}, & x = 0 \\ \frac{1}{6}, & x = 1 \\ \frac{1}{12}, & x = 2 \\ \frac{1}{20}, & x = 3 \\ \frac{1}{5}, & x = 4 \\ 0, & \text{otherwise} \end{cases}$$

Do notice that the difference of the sequential rounds can be expressed as

$$P(X = x) = \frac{1}{x} + \frac{1}{x+1} \text{ after } x+1 \text{ observations}$$

Which is what ChatGPT (2025) suggested we should do as our "distribution" to follow. ■

Revision for Solution

To find the expectation for the number of victories for player 1, the most straightforward approach would be to directly apply the definition of the expectation onto X , which would result in that:

$$E[X] = (0)(P = 0) + (1)(P = 1) + (2)P(X = 2) + (3)P(X = 3) + (4)P(X = 4) = \frac{77}{60}$$

But this would be a lame approach to find the expectation, and since the reader, hello Ariela, already pass the course of mathematical statistics why not find the expectation via the Moment Generating Function. Because we can clearly do it this way.

Subsection 4.2

Calculate $E[X]$

I invoke Paulo Mejia with the definition of Moment Generating Functions:

$$M_X(t) = E(e^{tX}) = \sum_{j=0}^4 e^{tj} P(X = j)$$

Then by the power of Nieves, I substitute each probability in its corresponding moment!

$$M_x(t) = \frac{1}{2}e^{0t} + \frac{1}{6}e^{1t} + \frac{1}{12}e^{2t} + \frac{1}{20}e^{3t} + \frac{1}{5}e^{4t}$$

With the blessing of Dorval, I derive $M_x(t)$ with respect to 0.

$$E[X] = M'_x(0) = \frac{1}{6}e^{1(0)} + \frac{2}{12}e^{2(0)} + \frac{3}{20}e^{3(0)} + \frac{4}{5}e^{4(0)}$$

Finally I finish with a MAC half lie half joke:

$$E[X] = \frac{77}{60}$$

Which evidently, by Nancy's omission of courtesy, both expectations are equivalent.

Problem 5:

A betting book recommends the following strategy for the roulette game. Bet one \$ 1 to the red color (lets define this as *rouge*. If the color red appears then take your winnings of \$ 1, and then leave. If the color red doesn't appear, and you lose your bet, then do more bets of \$ 1 for each red in the next to rounds of the roulette and then abandon the game (with or without any winnings).

Let X be the winnings of the player

The reader, hello Ariela, might be thinking "oh no another long discussion about a short probability problem." But I hope the reader, hello Ariela, is learning from reading this assignment. Because whats next is something that I believe might be useful in another course.

Revision for Solution

By how the strategy is structure, unfortunately one single distribution is not going to fit on the strategy. Instead what if we implement multiple distributions that can fit the strategy, kinda like if we compound them? Well reader, hello Ariela, have you heard of Compound or Mixture Distribution? Here is a short introduction to the subject: Siegrist, n.d.

So notice that the strategy can be separated into two scenarios:

- Scenario 1: The player bets \$ 1, and wins the first roulette by appearing a color red (with probability p), then they stop playing.
- Scenario 2: The player bets \$ 1, but red doesn't appear with probability $(1-p)$, the player consequentially loses \$ 1, but they then place two bets on red of \$ 1, each of these bets only have two outcomes, either the player wins or lose. Clearly both bets are independent of each other, and because the probability in each bet is the same, then we can define these bets as Binomially distributed. In fact, lets proposed that the outcomes for the scenario 2, the player's net Y is $Y \sim \text{Bin}(2, p)$, therefore express this scenario as the following:

1. if $Y = 0$, then the player has a net: -1
2. if $Y = 1$, then the player has a net: 0
3. if $Y = 2$, then the player has a net: 2

So if we mix both scenarios what we proceed with the overall distribution of the player's winnings X :

- probability p : $X = 1 + (k - k)$, here k is the initial bet which is \$ 1.

- probability $1 - p$: $X = Y - 1$, here $Y \sim \text{Bin}(2, p)$

Resulting in that the overall distribution X is:

$$X \sim \begin{cases} 1, & \text{with probability } p \\ Y - 1, Y \sim \text{Bin}(2, p) & \text{with probability } 1 - p \end{cases}$$

Subsection 5.1

Determine $P(\{X > 0\})$

Solution:

Using the mixture distribution defined above, notice that the second scenario only results in a positive net if and only if $Y = 2$. So the probability of the player getting a gain out of this game: $P(\{X > 0\})$. Which means, that events which contribute to this probabilities, are by construction, the first scenario (since the player will make a gain no matter what), however, as suggest before, on the second scenario, the player can only make a win, if $P(Y = 2)$. Therefore, because both scenarios are disjoint events, and by Axiom 3, then

$$P(\{X > 0\}) = P(\text{first scenario}) + P(\text{first bet lost AND } Y = 2)$$

Notice that on the second probability above, the event of losing first bet, and winning the next two upcoming bets are independent, then by the rule of independence:

$$P(\{X > 0\}) = P(\text{first scenario}) + P(\text{First bet lost}) \times P(Y = 2)$$

So by how the mixture distribution is defined, we then have that:

$$P(\{X > 0\}) = p + (1 - p)p^2 = p + p^2 - p^3$$

Here $P(Y = 2)$ results in p^2 , because Y is binomially distributed (Wikipedia, 2025a), then its PMF is

$$P(Y = y) = \binom{2}{y} p^y (1 - p)^{2-y}$$

If we let $y = 2$ then,

$$P(Y = 2) = \binom{2}{2} p^2 (1 - p)^0 = 1 \times p^2 \times 1 = p^2$$

Since we don't know the specific type of roulette that the strategy is playing with, we can't define the value of p , but suppose is a standard roulette, with 18 red slots, 18 black slots, and 2 green slots. Then $p = \frac{18}{38} = 0.47$, resulting in that

$$P(\{X > 0\}) = (0.47) + (0.47)^2 - (0.47)^3 = 0.58$$

■

Subsection 5.2

Are you convinced that the strategy is truly a winning Strategy? (Explain your reasoning.

Solution:

First lets determine what would be the expected net winning for each run the player plays.

By the definition of expectations:

$$E[X] = P(\text{win first bet}) \times 1 + P(\text{lose first bet}) \times E[Y - 1]$$

Then by the overall distribution X:

$$E[X] = p \times 1 + (1 - p) \times E[Y - 1]$$

Recall that $Y \sim \text{Bin}(2, p)$, so $E[Y] = 2p$ then by the linearity of the expectation,

$$E[Y - 1] = \left(\sum_i^2 p \right) - 1 = 2p - 1$$

Hence:

$$E[X] = p + (1 - p)(2p - 1) = p + 2p - 2p^2 - 1 + p = -1 + 4p - 2p^2$$

So if we substitute $p = 0.47$, then

$$E[X] = -1 + 4(0.47) - 2(0.47)^2 = 0.4382 \text{ (0.44 s.f)}$$

Clearly, in the player will make a small gain of 0.44 so it is relatively a good tactic. And I say tactic not strategy, this is because a strategy is played for the long run, while the tactics are short term or small actions that the player can take to make a small win.

The problem is that in the game of the roulette, there is no strategy that the player can consider to make an absolute winning due to the law of large numbers. Another way to look at it, if we consider the expectation of winning in the roulette game, where the strategy is placing your bet on the color red, then

$$E[W] = \frac{18}{38}(+1) \times \frac{20}{38} \times (-1) = \$ - 0.05$$

So regardless of how you strategize based on which tactics you implement, the overall game is expected that you eventually will return losing money (from your initial bet) rather than making any profit from your initial bet.

■

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